

23 Central Limit Theorem

- A **statistic** is a characteristic of the *sample* which can be computed from the data.
 - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
 - The probability distribution of a statistic is called a **sampling distribution**.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* and computing the value of the statistic for each sample.
 - The resulting standard deviation, called the **standard error**, measures the sample-to-sample variability of the statistic over many (hypothetical) samples of the same size.
 - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed.
- For many statistics (e.g., proportions, means, differences in means) — but not all —
 - Statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.
 - The pattern of sample-to-sample variability of the statistic follows, approximately, a Normal distribution

Example 23.1. In a previous example, we consider a discrete population with a population mean of $\mu = 72$ and a population standard deviation of $\sigma = 69.4$. Now we'll consider several other populations with a population mean of 72 and a population standard deviation of 69.4. We'll use an [applet](#) to simulate the sampling distribution of the sample mean $\bar{X}$ for samples of size n from several different populations. Enter 72 in the box for population mean, and 69.4 in the box for population SD.

1. Regardless of the shape of the population, what will $E(\bar{X})$ be? How we do interpret this?

2. Regardless of the shape of the population, what will $SD(\bar{X})$ be? How do we interpret this?
3. Choose a super-small sample size, like $n = 2$. Choose a population shape, and then run the simulation to generate many values of the sample mean. Then repeat for different population shapes. When the sample size is small, does the *shape* of the sample-to-sample distribution of sample means (plot on right) depend on the shape of the population (plot on the left)?
4. Now increase the sample size, and repeat for the several populations. What happens to the *shape* of the sample-to-sample distribution of sample means as the sample size increases?
5. Now choose a not-super-small sample size, like $n = 500$. Choose a population, and simulate *a single sample*. Look at the simulated sample (the middle plot), then repeat to simulate a few samples. Then repeat for the different populations. Does the *distribution of individual values within the sample* depend on the shape of the population?
6. When the sample size is large, does the *sample-to-sample distribution of sample means* depend on the shape of the population?

- For a representative random sample, the *distribution of individual values within the observed sample* should resemble the population distribution
 - Regardless of the sample size
- For many statistics (e.g., proportions, means, differences of means) — but not all — if the sample size is large enough the pattern of *sample-to-sample variability of values of the statistic* follows a Normal distribution
 - Regardless of the shape of the population distribution of individual values of the variable.
- The **Central Limit Theorem (CLT)** says that if n is large enough,

$\bar{X}_n$ has an approximate $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ distribution.

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution*.
- The above CLT is for a single population mean, but similar results apply for other statistics like sample proportions, differences in two sample means, differences in two sample proportions.

Example 23.2. Recall the population where every individual has an income of 10, 70, or 200 with probability 0.4, 0.4, 0.2. Now suppose that every individual has an income of 10, 70, 200, or 2000 with probability 0.4, 0.4, 0.19, 0.01. Simulate sample means for many samples of size $n = 30$ from this population. Is the distribution of sample means approximately Normal? What if $n = 100$?

- How large a sample size is required for normality of the sampling distribution of sample means does depend on the shape of the population distribution of individual values.
 - If the population distribution is Normal, then the sample-to-sample distribution of sample means is Normal for any sample size.
 - If the population distribution is “close to Normal” — e.g. symmetric, light tails — then smaller samples sizes are sufficient for the sample-to-sample distribution of sample means to be approximately Normal.
 - If the population distribution is “far from Normal” — e.g. severe skewness, heavy tails, extreme outliers — then larger sample sizes are required for the sample-to-sample distribution of sample means to be approximately Normal.

- You should certainly be aware that some population distributions require a large sample size for the CLT to kick in. However, in many situations the sample-to-sample distributions of sample means (or other statistics) is approximately Normal even for small or moderate sample sizes.

24 Confidence Intervals

- The estimators we have considered so far are *point* estimators; we estimate the population parameter θ , an unknown number, by the observed sample statistic $\hat{\theta}$, a number which can be computed based on sample data $X_1, \dots, X_n$
- For a particular sample a particular numerical value of $\hat{\theta}$ is observed, but there are many possible randomly selected samples and many possible values of the random variable $\hat{\theta}$.
- While for any particular sample we hope the observed value of $\hat{\theta}$ is close to the true value of θ , there will be some error in estimation due to natural sample-to-sample variability.
- Because of this natural sample-to-sample variability, it is usually better to provide a range of values when estimating a parameter
- In determining a plausible range of values, there is a trade-off between
 - Accuracy: the wider the interval the more confident we are that it actually contains the true parameter value
 - Precision: but the wider the interval the less precise of an estimate it provides
- The usual approach is to specify a desired degree of accuracy. Specifically, interval estimate procedures are constructed so that they will be accurate “most of the time”.
- A *confidence interval* estimates a population parameter based on sample data with
 - a plausible range of values and
 - a *confidence level* that the estimated range contains the true parameter value.
 - * By far, the most commonly used confidence level is 95%.
 - * But see Example 24.5 and related material about multiple confidence intervals
- A $1 - \alpha$ **confidence interval** for θ is an *interval estimator* of θ based on the sample $X_1, \dots, X_n$ of the form $[L(X_1, \dots, X_n), U(X_1, \dots, X_n)]$ which satisfies

$$P(L(X_1, \dots, X_n) \leq \theta \leq U(X_1, \dots, X_n)) = 1 - \alpha$$

- $L(X_1, \dots, X_n)$ and $U(X_1, \dots, X_n)$ represent, respectively, the lower and upper endpoints of the interval. L and U are *random variables* that vary from sample-to-sample.
- Typical values of α include 0.05 and 0.01. We express $1 - \alpha$ as a percentage. For example, $\alpha = 0.05$ corresponds to a 95% confidence interval.
- Often $L = \hat{\theta} - M$ and $U = \hat{\theta} + M$, where the **margin of error** M — where M can be a RV — accounts for both the sample-to-sample variability of $\hat{\theta}$ and the confidence level.
 - The margin of error is often of the form: multiple of standard error

- The multiple is determined by the confidence level and the distribution of the point estimator $\hat{\theta}$ (e.g., 2 for 95% confidence when the estimator follows a Normal distribution.)
 - * The larger the confidence level, the larger the margin of error — and the wider the range of plausible values — required to achieve that level of confidence.
- The **standard error** is the estimated sample-to-sample SD of the estimator, $SE(\hat{\theta}) = \widehat{SD}(\hat{\theta})$.
 - * The less variable the statistic (from sample to sample) the smaller the margin of error and the narrower the interval.
 - * In general, the larger the sample size, the smaller the SE of the statistic, the smaller the margin of error, and the narrower the interval.

Example 24.1. Consider the general problem of estimating a population mean μ based on a random sample of size n . A natural point estimator of μ is the sample mean $\bar{X}$. The central limit theorem says that, if n is large enough, $\bar{X}$ has an approximate Normal($\mu, \sigma/\sqrt{n}$) distribution, where σ is the population standard deviation. We'll use this theory to derive a confidence interval for μ .

1. Fill in the blank: for 95% of samples, the sample mean $\bar{X}$ is within [blank] of the population mean μ .

2. Fill in the blank: for 95% of samples, the interval with endpoints $\bar{X} \pm$ [blank] will contain the population mean μ .

3. The previous part suggests a formula for a CI for μ . However, this is not a practically useful formula; explain why. Then suggest how we can fix it.

4. Everything else being equal, how does the sample size affect the confidence interval and its margin of error: the larger the sample size...? Does this make sense?

5. Everything else being equal, how does the confidence level (e.g. 95%) affect the confidence interval and its margin of error: the larger the confidence level...? Does this make sense?

- A one-sample t confidence interval for a population mean μ has endpoints

$$\bar{X} \pm t^* \frac{S}{\sqrt{n}}$$

where S is the sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$ and t^* , the multiple corresponding to the confidence level, is from the t distribution with $n - 1$ degrees of freedom.

- For all practical purposes you can treat t distributions like Normal distributions and use the usual empirical rule
 - Technically, t^* is a little bigger than z^* to account for the additional sample-to-sample variability in S .
 - But in most situations encountered in practice, t^* and z^* are essentially the same.
- Assumptions:
 - Simple random sample from a large population, that is, independent observations
 - No bias in data collection
 - Population mean is the parameter you want to estimate

Confidence level	90%	95%	99%	99.7%	99.99%
Multiple ($z^* \approx t^*$)	1.7	2.0	2.6	3.0	4.0

Example 24.2. A frequently used example data set is the [Ames housing data set](#) which consists of residential properties sold in Ames, Iowa from 2006 to 2010. For more information about the variables in this data set, refer to the [data documentation](#).

Suppose we wish to use this data set to estimate μ , the *population mean sale price of single family homes*.

In the sample, there are 2425 single family homes with a sample mean sale price (\$K) of 185 and sample standard deviation 83.

1. Compute a 95% confidence interval.
2. Write a clearly worded sentence containing the conclusion of the confidence interval in context.
3. To what extent can you generalize your results? That is, what is the relevant “population”? What are some issues to consider?
4. Assuming that this is a random sample from the appropriate population, does the 95% confidence interval you computed in the previous part provide an accurate estimate of the corresponding parameter? That is, is μ contained within the CI? Explain.
5. Assuming that this is a random sample from the appropriate population, are we reasonably confident that μ is greater than 180K? Explain.
6. What if we wanted 95% confidence, but we wanted the margin of error to be no bigger than 1K. How could we achieve this?

- Because the value of the parameter is *unknown*, we never know for sure whether any one confidence interval contains the parameter or not.
- But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.
- Any value within the confidence interval is a plausible value of the parameter
- Values outside of the confidence interval are not plausible values of the parameter
- The CI could be wrong; we are confident that it is correct, but we’ll never know for sure
- There are two sources of differences between a statistic and a corresponding parameter
 - systematic errors due to **bias**
 - “chance errors” due to **sampling variability**
- While there is no bias inherent in simple random sampling from a population, bias can be present in other ways, including undercoverage, nonresponse, wording of questions, etc.
- Bias concerns the *center* of the sampling distribution of a statistic. A biased sampling procedure produces values which consistently overestimate or consistently underestimate the respective parameter.
- The inference procedures — confidence intervals and hypothesis tests — we will encounter assume that the data are collected in an unbiased way from a random sample.
- If there is bias present in data collection then the observed statistic (i.e. the center of the confidence interval) is likely to be a poor estimate of the parameter.
- The margin of error in a confidence interval only accounts for sampling variability, NOT bias.
- To see if bias is present — in which case the confidence interval cannot be trusted — ask: how was the data collected?

Example 24.3. For each of the following statements, identify if it is a valid interpretation of the confidence interval from Example 24.2. If not, explain why.

1. We estimate that 95% of single family homes in the population have sale price (\$K) between 181 and 188.
2. We estimate with 95% confidence that the sample mean sale price (\$K) of single family homes is between 181 and 188.

3. If many samples of this size were selected about 95% would produce a CI of (181, 188).

 4. In about 95% of samples of this size, the population mean sale price of single family homes will be between 181 and 188.

 5. If many hypothetical unbiased random samples of this size were selected, and a 95% confidence interval computed based on each sample, about 95% of these confidence intervals would contain the population mean sale price.
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- A CI is a conclusion about a population parameter — a number that *summarizes* the population (e.g. the population mean) — and not about individual values of a variable.
 - The conclusions from inference regard the *population* from which the random sample was selected; there is nothing to conclude for the sample.
 - No one particular sample (e.g. the one observed) or CI is necessarily good or representative of the many other possible samples or CIs.
 - The randomness is in the *sampling procedure*, which determines L and U , and not the parameter. The parameter θ is a fixed number. After selecting the sample, there is no randomness left. Either the CI contains the parameter or not; you just don't know which it is.
 - We are confident in our observed estimate because we are confident in our estimation *procedure*
 - By considering what would happen over many hypothetical random samples we are able to determine the necessary margin of error so that our CIs would contain the parameter in 95% of samples.
 - Because the value of the parameter is unknown, we never know for sure whether

any one confidence interval contains the parameter or not.

- But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.

Example 24.4. Continuing Example 24.2.

1. Consider the distribution of sale price. What shape do you expect this distribution to have? Why?
 2. A student says, “The distribution of sale price is clearly skewed to the right. Since it’s not Normal the confidence interval we computed, and the resulting conclusion about the population mean, is not valid.” Do you agree with this critique? Explain.
 3. Considering that the distribution of sale price is skewed to the right, suggest a valid criticism of the confidence interval we computed. Hint: what parameter is it estimating?
- The t interval for the population *mean* is valid as long as the sample size is large enough, regardless of the shape of the population distribution.
 - However, if the population distribution is highly skewed, then the population *median* might be a more appropriate parameter to consider as a measure of center.
 - Which is more appropriate, the population mean or the population median, depends on the purpose for the inference.
 - ALWAYS PLOT YOUR DATA, and consider the distribution of the variable. If the values in a representative sample are skewed or include large outliers, the same is also plausibly true for the population it’s supposed to represent.

Example 24.5. Suppose we wish to estimate population mean GPA of students at each of the 23 CSU campuses. For each campus, you select a random sample of students, collect the

GPA's of the students, and then use the sample data to compute a 95% confidence interval. That is you compute 23 separate 95% confidence intervals, based on 23 independent random samples, for each of 23 population means.

1. First just consider Cal Poly. What is the probability that your confidence interval based on the Cal Poly sample contains the true population mean GPA at Cal Poly?

2. What is the probability that all 23 of your 95% confidence intervals contain their respective population mean?

3. How confident are you that all 23 of your 95% confidence intervals contain their respective population mean?

- When computing multiple confidence intervals, the multiple (z^* or t^*) of each confidence interval should be increased to account for the **issue of multiple comparisons**.
 - For example, with a multiple of 2 in a *single* confidence interval, we are 95% confident that the interval contains the corresponding parameter.
 - However, if we use a multiple of 2 for each of 10 confidence intervals, we are only 60% confident¹ that *all* of the 10 95% confidence intervals contain their respective parameters.
- A **Bonferroni** adjustment, splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the z^*/t^* multiple for 99.5% confidence when constructing each interval.
 - A Bonferroni adjustment guarantees **simultaneous confidence**

¹Of course there is also lots of potential for bias in data collection, but we always assume we have an unbiased random sample. Even when the data are a random sample from the population, there can still be bias in estimation.

- But it is conservative, and tends to produce wider intervals than necessary
 - There are other methods; for example, the *Tukey-Kramer method* is often used for *all pairwise comparisons of means* (e.g., in ANOVA)
4. Suppose we want simultaneous 95% confidence that all 23 CIs contain their respective mean. Find the z^* multiple corresponding to a Bonferroni adjustment. How many times wider will each CI be than the unadjusted intervals?
 5. How confident are you that all 23 of your adjusted confidence intervals contain their respective population mean?

Example 24.6. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes (including condos, townhouses, etc). In particular, suppose we wish to estimate $\mu_S - \mu_N$, the difference in population mean sale price between single family homes and non-single family homes.

1. Suggest a point estimator, compute its variance, and suggest a formula for a CI for $\mu_S - \mu_N$.

2. The table below summarizes the sample data. Compute a 95% confidence interval.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

3. Write a clearly worded sentence interpreting the confidence interval from the previous part in context.

4. Are we reasonably confident that single family homes tend to have higher sale prices than other homes? Explain.

- For many statistics (e.g. means, differences in means, proportions, differences in proportions) the sample-to-sample pattern of variability of values of the statistic over many unbiased random samples has an approximate Normal shape centered at the true value of the population parameter

- In such situations confidence intervals take the form

$$\text{statistic} \pm \text{multiple of its SE}$$

- If two samples are selected *independently* from each of two populations with parameters θ_1 and θ_2 , then an approximate $1 - \alpha$ CI for $\theta_1 - \theta_2$ has endpoints

$$\hat{\theta}_1 - \hat{\theta}_2 \pm t^* \text{SE}(\hat{\theta}_1 - \hat{\theta}_2)$$

where

$$\text{SE}(\hat{\theta}_1 - \hat{\theta}_2) = \sqrt{\text{SE}(\hat{\theta}_1)^2 + \text{SE}(\hat{\theta}_2)^2}$$

- The above assumes that $\hat{\theta}_1$ and $\hat{\theta}_2$ each have an approximate Normal distribution, and so, since they are independent, $\hat{\theta}_1 - \hat{\theta}_2$ also has an approximate Normal distribution.
- A difference provides an absolute comparison between groups.
 - Ratios can be used to provide relative comparisons; however, the CI formulas are more complicated, but we'll see another approach soon

25 Bootstrap Confidence Intervals

The value of any statistic varies from sample to sample. Statistical inference procedures such as confidence intervals are based on sampling distributions which describe how the values of a statistic vary from sample to sample over many possible samples. In particular, the standard error of a statistic measures its degree of sample-to-sample variability.

If a population distribution is assumed, then we can approximate the sampling distribution of a statistic and characteristics like SE via simulation. But what if the population distribution is completely unknown? For some statistics — most notably means, proportions and certain transformations of them — the Central Limit Theorems and its relatives (e.g., Delta Method) guarantee that the sampling distribution will be approximately Normal for large samples, regardless of the population distribution. However, not every statistic has an approximate Normal distribution, so we need other methods for approximating sampling distributions. But how do we estimate the sampling distribution of a statistic based only on sample data? Bootstrapping provides a simple but extremely versatile and powerful method for doing so.

Example 25.1. For purposes of illustration, we'll start with a small data set. The table below contains data on all the quidditch matches (for which the final score was discernible) in the *Harry Potter* books¹. The sample mean is $\bar{x} = 340$ and the sample SD is $\hat{\sigma} = 134.8$ points². Say we want a confidence interval for μ , the population mean quidditch score, based on this sample.

200	230	250	260	380	470	590
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1. Warmup exercise: If we assume that quidditch scores follow a $N(345, 140)$ distribution, explain how you would use simulation to approximate the sampling distribution of the sample mean. Note: we really want to approximate the sampling distribution without assuming anything about the population distribution; this is just to remind you of how we have done things so far. Also note: the values 345 and 140 are just made up; we're assuming three things: (1) scores follow a Normal distribution, (2) population mean is 345, (3) population SD is 140.

¹Source: http://harrypotter.wikia.com/wiki/Inter-House_Quidditch_Cup

²Using the $1/n$ version of sample SD here for reasons that we'll see later. Spoile alert: in bootstrapping, we treat the sample as if were the population, so we're treating the sample mean as if it were the population mean, so we really do want the average distance from the sample mean without "unbiasing" it (which is what dividing by $n - 1$ instead of n does.)

2. Warmup exercise: How could you estimate the SE of the sample mean based on the above simulation? Compare to the theoretical formula.
3. Now suppose we make no assumptions about the population distribution. How could you estimate the population distribution based only on the observed sample data? Construct a spinner corresponding to this distribution.
4. Continuing the previous part, explain how you would use simulation to approximate the sampling distribution of the sample mean *without knowing the population distribution*.
5. Suppose that instead of using a spinner, you wrote each observed value on a piece of paper. How would you use the pieces of paper to simulate a sample?
6. Conduct by hand a few repetitions of the simulation from the previous part.
7. Use software to perform the simulation. How would you use the simulation results to approximate the SE? Compare to the usual formula; hint: what plays the role of

“population” SD in this simulation?

8. How could you find a confidence interval for the population mean?

- The sampling distribution of a statistic can be approximated by simulating many samples *from the population* and computing the value of the statistic for each simulated sample.
- The idea behind **bootstrapping** is that if the sample data is representative of the population, then we can approximate the process of selecting a sample from the population by *treating the observed sample as if it were the population and selecting a sample — called a resample or bootstrap sample — from the observed sample.*
- A **bootstrap sample** *resamples* the data from the *observed sample*, drawing the same number of observations, but *with replacement*.
- The **bootstrap distribution** of a statistic is obtained by simulating many bootstrap samples and computing the value of the statistic for each sample
 - Simulate a (re)samples of size n with replacement from the observed sample
 - Compute the value of the statistic for the simulated bootstrap sample
 - Repeat many times, recording the value of the statistic for each repetition
- The **bootstrap standard error** of a statistic is the standard deviation of the bootstrap distribution of the statistic.
- The bootstrap distribution of a statistic approximates some, but not all, features of the sampling distribution of the statistic.
 - The shape of the bootstrap distribution usually approximates the shape of the sampling distribution
 - The variability of the bootstrap distribution usually approximates the variability of the sampling distribution. That is, the bootstrap SE usually approximates the true SE.
 - The center of the bootstrap distribution does NOT accurately approximate the center of the sampling distribution
 - The bootstrap estimate of bias usually approximates the bias of the statistic.
- There are many methods for constructing bootstrap confidence intervals for parameters; we’ll only study a few.

- **Bootstrap Normal interval:** If the bootstrap distribution for the corresponding statistic is approximately Normal, then an approximate 95% confidence interval has endpoints

$$\text{observed statistic} \pm 2 \times \text{bootstrap SE}$$

- **Bootstrap percentile interval:** The 95% bootstrap percentile interval is the interval with endpoints equal to the 2.5th and 97.5th percentiles of the bootstrap distribution.
- One of the main advantages of bootstrapping is that it can be used to approximate the SE of *any statistic*.

Example 25.2. Continuing the previous exercise, say we want to estimate the population *median* quidditch score.

1. Describe in detail how would use simulation to approximate the bootstrap distribution. Then conduct the simulation. What are the possible values in the bootstrap distribution? Why?
2. Which bootstrap confidence interval would be more appropriate, Normal or percentile? Find and interpret the appropriate confidence interval.

Example 25.3. Using the Ames housing data set, suppose we wish to estimate the population *median* sale price of single family homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the population median.

2. Conduct the simulation and find and interpret the confidence interval.

Example 25.4. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population means*.
2. Conduct the simulation and find and interpret the confidence interval.
3. Describe in detail how you would use bootstrapping to find a confidence interval for the *ratio of population medians*.
4. Conduct the simulation and find and interpret the confidence interval.

Example 25.5. Continuing with the Ames housing data. Suppose we want to estimate the *correlation* between sale price and square footage (“Gr Liv Area”).

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *population correlation coefficient* between sale price and square footage for single family

homes.

2. Conduct the simulation and find and interpret the confidence interval.
 3. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population correlation coefficients* between sale price and square footage for single family homes and other homes.
 4. Conduct the simulation and find and interpret the confidence interval.
- The accuracy of bootstrap procedures is primarily influenced by the size of the original sample.
 - Bootstrapping methods do not overcome the weaknesses of small samples.

Example 25.6. Consider again the quidditch data

200 230 250 260 380 470 590

1. What is the center of the *sampling distribution* of the sample mean? Why? Given a single observed sample, what is the center of the *bootstrap distribution* of the sample mean?

2. Suppose we want to estimate the population SD? Is the sample SD an unbiased estimator of the population SD? Does this cause any issues for CIs?

3. How could we use bootstrapping to estimate the bias of the sample SD (as an estimator of the population SD)?
 - The bootstrap distribution of a statistic approximates many, but not all, features of the sampling distribution of the statistic.
 - In particular, the center of the bootstrap distribution will generally not be the same as the center of the sampling distribution.
 - This can lead to coverage problems (e.g., less than the stated 95%) for bootstrap percentile intervals, especially for statistics which are biased estimators of their parameters.
 - Bootstrap percentile intervals do not fully account for differences between the observed statistic and the true parameter
 - Remember, there are two sources of differences between a statistic and a corresponding parameter
 - systematic errors due to *estimation bias*³
 - “chance errors” due to natural *sample-to-sample variability*
 - The usual confidence procedures assume that there is no bias in estimation. The margin of error of a confidence interval typically only accounts for natural sample-to-sample variability. So if there is bias in estimation, the actual coverage probability of the confidence interval procedure might be less than the nominal confidence level.
 - Recall that the bias of an estimator $\hat{\theta}$ of a parameter θ is $E(\hat{\theta}) - \theta$
 - That is, the bias of an estimator is the difference between the center of its sampling distribution and the true (but unknown) value of the parameter
 - If a statistic is *unbiased*, the center of its sampling distribution should be equal to the true (but unknown) value of the parameter.

³Of course there is also lots of potential for bias in data collection, but we always assume we have an unbiased random sample. Even when the data are a random sample from the population, there can still be bias in estimation.

- lower endpoint = $2 \times \text{observed statistic} - 97.5\text{th percentile of bootstrap distribution}$
upper endpoint = $2 \times \text{observed statistic} - 2.5\text{th percentile of bootstrap distribution}$

- 180